Seminar 3:

Classroom problems and worked solutions

25117 - Econometrics

Universitat Pompeu Fabra

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The issue: what do study hours tell us about scores?

- Fit a line to study hours and test scores, then calculate what the fitted line leaves unexplained.
- Assess what an estimated slope says about prediction and what additional assumptions support a causal interpretation.
- Measure the uncertainty in both coefficients and use it to test claims about the population line.

Classroom problems

Show the formula before substituting numbers. Worked solutions follow each problem, with answer numbers matching the subquestions.

Problem 1: What does the population regression predict?

In a population model, Y is a student's test score in points and X is weekly study time in hours, with support $X \in \{1, 2, 3, 4, 5\}$:

$$Y = 50 + 2X + U, \qquad \mathbb{E}(U \mid X) = 0, \qquad \mathbb{E}(Y \mid X = x) = 50 + 2x.$$

1. Identify the outcome, regressor, intercept, and slope. State the units of the intercept and slope.
2. Calculate the conditional mean score at 2 and 4 study hours. What change in the conditional mean accompanies this change in hours?
3. A student studies 3 hours and scores 59. Find this student's realized population error. Explain why the individual score can differ from the conditional mean even though $\mathbb{E}(U \mid X) = 0$.

Solution 1: Conditional means and an individual error

1. Y is the outcome in test points; X is the regressor in weekly study hours. The intercept $\beta_0 = 50$ is in points and the slope $\beta_1 = 2$ is in points per hour. The line's intercept at $x = 0$ lies outside the given support.

2. Using $\mathbb{E}(Y \mid X = x) = \beta_0 + \beta_1 x$,

$$\mathbb{E}(Y \mid X = 2) = 50 + 2(2) = 54, \quad \mathbb{E}(Y \mid X = 4) = 50 + 2(4) = 58.$$

$$\Delta\mathbb{E}(Y \mid X) = \beta_1 \Delta x = 2(4 - 2) = \boxed{4 \text{ points}}.$$

3. The realized population error is the distance from the population line:

$$u = y - (\beta_0 + \beta_1 x) = 59 - [50 + 2(3)] = 59 - 56 = \boxed{3 \text{ points}}.$$

At $X = 3$, the conditional mean is 56 and this student's score is 59. The condition $\mathbb{E}(U \mid X = 3) = 0$ concerns the average error among students at that value of X; individual errors can be positive or negative.

Problem 2: Which line best fits five students?

For a new sample, weekly study hours X and test scores Y are

Student i	1	2	3	4	5
x_i (hours)	1	2	3	4	5
y_i (points)	52	54	53	58	58

1. Write the OLS objective $Q(a, b)$ for a candidate line $a + bx$. Differentiate with respect to a and b to obtain the two first-order conditions.
2. Calculate $\bar{x}$, $\bar{y}$, and a table of $x_i - \bar{x}$, $y_i - \bar{y}$, their cross-products, and $(x_i - \bar{x})^2$.
3. Derive $\widehat{\beta}_0 = \bar{y} - \widehat{\beta}_1 \bar{x}$ from the first condition. Calculate the OLS slope and intercept, showing why the slope also equals s_{XY}/s_X^2.

Solution 2: The objective and the normal equations

1. For any candidate intercept a and slope b, OLS minimizes

$$Q(a, b) = \sum_{i=1}^{n} (y_i - a - bx_i)^2.$$

Differentiating gives

$$\frac{\partial Q}{\partial a} = -2 \sum_{i=1}^{n} (y_i - a - bx_i), \qquad \frac{\partial Q}{\partial b} = -2 \sum_{i=1}^{n} x_i (y_i - a - bx_i).$$

At $(a, b) = (\widehat{\beta}_0, \widehat{\beta}_1)$, set both derivatives to zero:

$$\sum_i (y_i - \widehat{\beta}_0 - \widehat{\beta}_1 x_i) = 0, \qquad \sum_i x_i (y_i - \widehat{\beta}_0 - \widehat{\beta}_1 x_i) = 0.$$

Solution 2: Centered observations and OLS coefficients (continued)

2. The means are

$$\bar{x} = \frac{1}{n}\sum_i x_i = \frac{15}{5} = 3, \qquad \bar{y} = \frac{1}{n}\sum_i y_i = \frac{275}{5} = 55.$$

i	x_i	y_i	$x_i - \bar{x}$	$y_i - \bar{y}$	$(x_i - \bar{x})(y_i - \bar{y})$	$(x_i - \bar{x})^2$
1	1	52	-2	-3	6	4
2	2	54	-1	-1	1	1
3	3	53	0	-2	0	0
4	4	58	1	3	3	1
5	5	58	2	3	6	4
Sum	15	275	0	0	16	10

Solution 2: The slope and intercept (continued)

3. The first normal equation implies

$$n\bar{y} - n\widehat{\beta}_0 - n\widehat{\beta}_1\bar{x} = 0 \quad \Longrightarrow \quad \boxed{\widehat{\beta}_0 = \bar{y} - \widehat{\beta}_1\bar{x}}.$$

Combining the two normal equations yields

$$\sum_i (x_i - \bar{x})(y_i - \bar{y}) = \widehat{\beta}_1 \sum_i (x_i - \bar{x})^2.$$

The common divisor $n - 1$ cancels from the covariance-to-variance ratio:

$$\widehat{\beta}_1 = \frac{\sum_i (x_i - \bar{x})(y_i - \bar{y})}{\sum_i (x_i - \bar{x})^2} = \frac{s_{XY}}{s_X^2} = \frac{16/(5-1)}{10/(5-1)} = \frac{16}{10} = \boxed{1.6}.$$

$$\widehat{\beta}_0 = \bar{y} - \widehat{\beta}_1\bar{x} = 55 - 1.6(3) = \boxed{50.2}, \qquad \widehat{y} = 50.2 + 1.6x.$$

Problem 3: What does OLS leave unexplained?

Use the five observations and estimated line from Problem 2:

$$\widehat{y}_i = 50.2 + 1.6x_i.$$

1. Calculate every fitted value and residual $\widehat{u}_i = y_i - \widehat{y}_i$. Sketch the observations and fitted line, and identify a residual as a vertical distance.
2. Verify that $\sum_i \widehat{u}_i = 0$ and $\sum_i x_i \widehat{u}_i = 0$. Check that the fitted line passes through $(\overline{x}, \overline{y})$.
3. Calculate the sum of squared residuals, $SSR = \sum_i \widehat{u}_i^2$. Compare it with the squared-error sums for the constant prediction 55 and the candidate line $50 + 2x$.

Solution 3: Fitted scores and vertical residuals

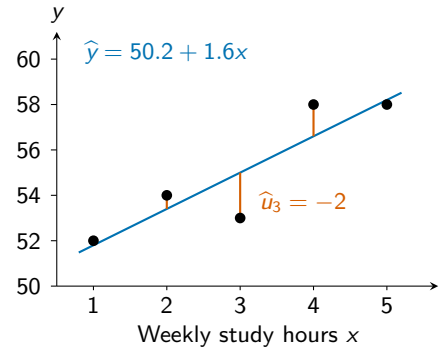
1. Apply the fitted-value and residual formulas to each observation:

$$\widehat{y}_i = \widehat{\beta}_0 + \widehat{\beta}_1 x_i = 50.2 + 1.6 x_i, \qquad \widehat{u}_i = y_i - \widehat{y}_i.$$

x_i	y_i	$\widehat{y}_i$	$\widehat{u}_i$
1	52	51.8	.2
2	54	53.4	.6
3	53	55.0	−2.0
4	58	56.6	1.4
5	58	58.2	−.2

For student 3,

$$\widehat{u}_3 = 53 - [50.2 + 1.6(3)] = -2.$$



Residuals use the estimated sample line. Population errors use the true population coefficients.

Solution 3: Normal equations and squared errors (continued)

2. The normal equations are satisfied:

$$\sum_i \widehat{u}_i = .2 + .6 - 2 + 1.4 - .2 = 0,$$

$$\sum_i x_i \widehat{u}_i = 1(.2) + 2(.6) + 3(-2) + 4(1.4) + 5(-.2) = 0.$$

The line passes through the sample means:

$$\widehat{y}(\overline{x}) = \widehat{\beta}_0 + \widehat{\beta}_1 \overline{x} = 50.2 + 1.6(3) = 55 = \overline{y}.$$

3. The OLS residual sum of squares is

$$SSR = \sum_i \widehat{u}_i^2 = .2^2 + .6^2 + (-2)^2 + 1.4^2 + (-.2)^2 = \boxed{6.4}.$$

Prediction	Residuals	Sum of squared errors
Constant 55	$-3, -1, -2, 3, 3$	32
Candidate $50 + 2x$	$0, 0, -3, 0, -2$	13
OLS $50.2 + 1.6x$	$.2, .6, -2, 1.4, -.2$	6.4

OLS has lower squared error than either alternative. These sample fitting identities do not establish causality.

Problem 4: What does a 10-percentage-point increase predict?

An estimated district regression relates the average test score Y to the share p of students receiving subsidized meals:

$$\widehat{y} = 850 - 150p.$$

The observed shares range from .20 to .80. The standard deviation of district scores is 60 points.

1. Predict scores at $p = .40$ and $p = .50$. Calculate the predicted change for this 10-percentage-point increase in the share.
2. Express the regressor as a percentage, $P = 100p$. Rewrite the fitted line and state the slope's new units. Check that the same change still gives the same predicted score difference.
3. What does the line predict at $p = 0$, and why does interpreting that value require extrapolation? Express the 15-point predicted decrease as a fraction of the outcome's standard deviation.

Solution 4: Shares, percentage points, and predicted scores

1. For a share p, the fitted score is $\widehat{y}(p) = 850 - 150p$:

$$\widehat{y}(.40) = 850 - 150(.40) = 790, \qquad \widehat{y}(.50) = 850 - 150(.50) = 775.$$

$$\Delta\widehat{y} = \widehat{\beta}_1 \Delta p = -150(.50 - .40) = \boxed{-15 \text{ points}}.$$

A one-unit increase in the share means 100 percentage points. Here $\Delta p = .10$ means 10 percentage points.

2. Since $P = 100p$, substitute $p = P/100$:

$$\widehat{y} = 850 - 150\left(\frac{P}{100}\right) = \boxed{850 - 1.5P}.$$

The slope is now -1.5 test-score points per percentage point. In these units,

$$\Delta\widehat{y} = (-1.5)(50 - 40) = \boxed{-15 \text{ points}}.$$

Changing the units changes the coefficient's numerical scale and preserves the predictions.

Solution 4: The intercept and a standard-deviation benchmark (continued)

3. The line's value at zero share is

$$\widehat{y}(0) = 850 - 150(0) = \boxed{850 \text{ points}}.$$

But $p = 0$ is outside the observed range $[.20, .80]$. Applying the fitted line there extrapolates beyond the available data; the intercept is not evidence about districts observed near zero share.

The predicted change relative to the outcome's standard deviation is

$$\frac{\Delta\widehat{y}}{s_Y} = \frac{-15}{60} = \boxed{-.25}.$$

The 10-percentage-point increase in meal share is associated with a predicted score decrease of **one quarter of a standard deviation**.

Interpretation

The estimated line describes an association. A causal interpretation requires evidence about how meal shares and other determinants of scores are related.

Problem 5: What does a tutoring indicator estimate?

Let $D = 1$ indicate tutoring participation and $D = 0$ no tutoring. Six students have the following test scores:

Group	Scores y_i (points)		
$D = 0$	60	63	66
$D = 1$	67	70	73

Fit $\widehat{y}_i = \widehat{\beta}_0 + \widehat{\beta}_1 D_i$. No assignment mechanism is specified.

1. Calculate each group's mean score, $\overline{y}_0$ and $\overline{y}_1$.
2. Split the OLS objective into sums for the two groups. Use the first-order conditions to explain why the intercept is $\overline{y}_0$ and the slope is $\overline{y}_1 - \overline{y}_0$.
3. Calculate the fitted scores, residuals, and total *SSR*.
4. Does the slope identify a causal tutoring effect from these data alone? State a random-assignment design under which the difference could estimate an average causal effect.

Solution 5: A binary slope is a difference in group means

1. For group d, $\bar{y}_d = n_d^{-1} \sum_{i:D_i=d} y_i$. Here,

$$\bar{y}_0 = \frac{60+63+66}{3} = 63, \qquad \bar{y}_1 = \frac{67+70+73}{3} = 70.$$

2. For candidate coefficients (a, b), write

$$Q(a,b) = \sum_{i:D_i=0} (y_i - a)^2 + \sum_{i:D_i=1} (y_i - a - b)^2.$$

The first-order condition for b gives

$$\frac{\partial Q}{\partial b} = -2 \sum_{i:D_i=1} (y_i - a - b) = 0 \quad \Longrightarrow \quad a + b = \bar{y}_1.$$

The first-order condition for a then gives

$$\frac{\partial Q}{\partial a} = -2 \sum_{i:D_i=0} (y_i - a) - 2 \underbrace{\sum_{i:D_i=1} (y_i - a - b)}_{=0} = 0 \quad \Longrightarrow \quad a = \bar{y}_0.$$

Thus

$$\widehat{\beta}_0 = \bar{y}_0 = \boxed{63}, \qquad \widehat{\beta}_1 = \bar{y}_1 - \bar{y}_0 = 70 - 63 = \boxed{7}.$$

Solution 5: Group predictions and a causal interpretation (continued)

3. Use $\widehat{y}_i = 63 + 7D_i$ and $\widehat{u}_i = y_i - \widehat{y}_i$:

Group	Observed scores	Fitted scores	Residuals
$D = 0$	60, 63, 66	63, 63, 63	-3, 0, 3
$D = 1$	67, 70, 73	70, 70, 70	-3, 0, 3

$$SSR = \sum_i \widehat{u}_i^2 = [(-3)^2 + 0^2 + 3^2] + [(-3)^2 + 0^2 + 3^2] = 18 + 18 = \boxed{36 \text{ points}^2}.$$

4. With no assignment mechanism specified, the estimated slope describes a **7-point sample difference** between participants and nonparticipants.

If students were individually randomized to tutoring, with full take-up, no spillovers, and complete posttest follow-up, this difference could estimate the average causal effect for the study population.

Zero average residuals within groups are an OLS fitting property; they do not establish those design conditions.

Problem 6: How much score variation does the line explain?

Use the five observations and fitted values from Problems 2–3:

x_i	1	2	3	4	5
y_i	52	54	53	58	58
$\widehat{y}_i$	51.8	53.4	55	56.6	58.2
$\widehat{u}_i$	.2	.6	-2	1.4	$-.2$

The sample mean is $\bar{y} = 55$.

1. Compute the total sum of squares (TSS), explained sum of squares (ESS), and residual sum of squares (SSR). Verify $TSS = ESS + SSR$.
2. Calculate R^2 using both ESS/TSS and $1 - SSR/TSS$.
3. Calculate the sample variances of Y and $\widehat{Y}$. Relate their ratio to R^2 and interpret the result.

Solution 6: Decomposing the observed score variation

1. For OLS with an intercept, $\overline{\widehat{y}} = \overline{y} = 55$. Define

$$TSS = \sum_i (y_i - \overline{y})^2, \quad ESS = \sum_i (\widehat{y}_i - \overline{y})^2, \quad SSR = \sum_i \widehat{u}_i^2.$$

Student i	$(y_i - 55)^2$	$(\widehat{y}_i - 55)^2$	$\widehat{u}_i^2$
1	9	10.24	.04
2	1	2.56	.36
3	4	0	4
4	9	2.56	1.96
5	9	10.24	.04
Sum	$TSS = 32$	$ESS = 25.6$	$SSR = 6.4$

Thus $TSS = ESS + SSR$: $32 = 25.6 + 6.4$.

2. Both expressions give

$$R^2 = \frac{ESS}{TSS} = \frac{25.6}{32} = \boxed{.8}, \qquad R^2 = 1 - \frac{SSR}{TSS} = 1 - \frac{6.4}{32} = \boxed{.8}.$$

Solution 6: The variance ratio and its interpretation

3. Both sample variances use the same divisor $n-1$:

$$s_Y^2 = \frac{\sum_i (y_i - \bar{y})^2}{n-1} = \frac{32}{5-1} = 8 \text{ points}^2,$$

$$s_{\hat{Y}}^2 = \frac{\sum_i (\hat{y}_i - \bar{\hat{y}})^2}{n-1} = \frac{25.6}{5-1} = 6.4 \text{ points}^2.$$

Since the fitted values have mean $\bar{y}$,

$$\frac{s_{\hat{Y}}^2}{s_Y^2} = \frac{ESS/(n-1)}{TSS/(n-1)} = \frac{ESS}{TSS} = \frac{6.4}{8} = \boxed{.8 = R^2}.$$

The fitted line accounts for 80% of the observed variation in scores around their sample mean. The remaining 20% is in the residuals.

This describes fit in these five observations. It does not by itself establish accuracy on future data or a causal effect of study hours.

Problem 7: How large are the residuals around the line?

Use Problems 2–3: $n = 5$, $\widehat{y} = 50.2 + 1.6x$, $SSR = 6.4$, and $TSS = 32$.

1. Calculate SER^2 and SER. Compare SER with the sample standard deviation s_Y of scores.
2. Calculate the in-sample root mean squared error, $RMSE$. Explain why its denominator differs from that of SER.
3. Predict the score at $x = 3.5$ study hours and calculate $\widehat{y} \pm SER$. Is this automatically a 95% prediction interval for a new student's score? Explain.

Solution 7: SER, RMSE, and their denominators

1. Two fitted coefficients leave $n-2$ residual degrees of freedom:

$$SER^2 = \frac{SSR}{n-2} = \frac{6.4}{5-2} = \frac{32}{15} \approx 2.1333 \text{ points}^2,$$

$$SER = \sqrt{SER^2} = \sqrt{\frac{32}{15}} \approx \boxed{1.4606 \text{ points}}.$$

For comparison,

$$s_Y = \sqrt{\frac{TSS}{n-1}} = \sqrt{\frac{32}{4}} = \sqrt{8} \approx 2.8284 \text{ points}.$$

SER measures residual spread around the line; s_Y measures spread around the sample mean.

2. The in-sample average squared error divides by all n observations:

$$RMSE = \sqrt{\frac{1}{n}\sum_i \widehat{u}_i^2} = \sqrt{\frac{SSR}{n}} = \sqrt{\frac{6.4}{5}} = \sqrt{1.28} \approx \boxed{1.1314 \text{ points}}.$$

Thus $SER > RMSE$: their squared numerators agree, but their divisors are 3 and 5. Neither quantity is the standard error of a regression coefficient.

Solution 7: Predicting at three and a half study hours

3. Insert $x = 3.5$ into the fitted line:

$$\widehat{y}(x) = \widehat{\beta}_0 + \widehat{\beta}_1 x, \qquad \widehat{y}(3.5) = 50.2 + 1.6(3.5) = \boxed{55.8 \text{ points}}.$$

A band extending one *SER* on either side is

$$\widehat{y}(3.5) \pm \mathit{SER} = 55.8 \pm \sqrt{32/15} \approx \boxed{[54.339,\ 57.261]} \text{ points.}$$

This is a descriptive one-residual-standard-deviation band. It does not automatically have 95% coverage for a new student's score.

A formal prediction interval needs a suitable error-distribution model and critical value. It must also account for uncertainty in the new student's error and the estimated regression coefficients.

Problem 8: Can the best line miss the conditional mean?

A finite population has $X \in \{-1, 0, 1\}$, each with probability $1/3$, and $Y = X^2$. The population best linear predictor minimizes

$$Q(a, b) = \mathbb{E}[(Y - a - bX)^2].$$

Denote its coefficients by β_0^* and β_1^*.

1. Compute $\mathbb{E}(X)$, $\mathbb{E}(Y)$, $\mathrm{Var}(X)$, and $\mathrm{Cov}(X, Y)$. Use the population first-order conditions to find β_0^* and β_1^*.
2. Compare the best linear predictor $\beta_0^* + \beta_1^* x$ with the true conditional mean $\mathbb{E}(Y \mid X = x)$ at all three values of x.
3. Define the population projection error $V = Y - \beta_0^* - \beta_1^* X$. Verify $\mathbb{E}(V) = \mathbb{E}(XV) = 0$. Is $\mathbb{E}(V \mid X) = 0$ also true?

Solution 8: Finding the population best line

1. Using equal probabilities,

$$\mathbb{E}(X) = \frac{-1+0+1}{3} = 0, \qquad \mathbb{E}(Y) = \mathbb{E}(X^2) = \frac{1+0+1}{3} = \frac{2}{3},$$

$$\text{Var}(X) = \mathbb{E}(X^2) - [\mathbb{E}(X)]^2 = \frac{2}{3}, \qquad \text{Cov}(X, Y) = \mathbb{E}(XY) - \mathbb{E}(X)\mathbb{E}(Y) = \mathbb{E}(X^3) = 0.$$

The population first-order conditions for $Q(a, b)$ are

$$\frac{\partial Q}{\partial a} = -2\mathbb{E}(Y - a - bX) = 0, \qquad \frac{\partial Q}{\partial b} = -2\mathbb{E}[X(Y - a - bX)] = 0.$$

The first gives $a = \mathbb{E}(Y) - b\mathbb{E}(X)$. Substituting into the second gives

$$\text{Cov}(X, Y) - b\,\text{Var}(X) = 0 \quad \Longrightarrow \quad \beta_1^* = \frac{\text{Cov}(X, Y)}{\text{Var}(X)} = 0,$$

$$\beta_0^* = \mathbb{E}(Y) - \beta_1^*\mathbb{E}(X) = \boxed{\frac{2}{3}}.$$

These are population projection coefficients, rather than sample estimates.

Solution 8: Orthogonality and conditional means

2. Since $Y = X^2$ deterministically, $\mathbb{E}(Y \mid X = x) = x^2$.

x	-1	0	1
True conditional mean x^2	1	0	1
Best linear predictor $\beta_0^* + \beta_1^* x$	$2/3$	$2/3$	$2/3$
Projection error V	$1/3$	$-2/3$	$1/3$

3. The projection error is $V = Y - 2/3$. Therefore

$$\mathbb{E}(V) = \frac{1/3 - 2/3 + 1/3}{3} = 0, \qquad \mathbb{E}(XV) = \frac{-1/3 + 0 + 1/3}{3} = 0.$$

However,

$$\mathbb{E}(V \mid X = x) = x^2 - \frac{2}{3} \neq 0 \quad \text{for every } x \in \{-1, 0, 1\}.$$

The error is uncorrelated with X, but its conditional mean is nonzero. Zero covariance is weaker than zero conditional mean; this example concerns prediction, not a causal effect.

Problem 9: Can disadvantage reverse the observed tutoring gap?

Suppose the structural model for test scores is

$$Y = 80 + 5D - 10Z + V,$$

where D indicates tutoring participation and Z measures disadvantage. Assume finite second moments. In this model, changing D from 0 to 1 while holding Z and V fixed raises the score by 5 points. Assume

$$\mathbb{E}(V \mid D) = 0, \quad \mathbb{E}(Z \mid D = 0) = 1, \quad \mathbb{E}(Z \mid D = 1) = 2, \quad \Pr(D = 1) = .5.$$

1. Calculate $\mathbb{E}(Y \mid D = 0)$, $\mathbb{E}(Y \mid D = 1)$, and the observed mean gap.
2. If Z is omitted, define the structural error $U = -10Z + V$. Compute $\mathbb{E}(U \mid D = 0)$ and $\mathbb{E}(U \mid D = 1)$. Does zero conditional mean hold?
3. Find $\mathrm{Var}(D)$ and $\mathrm{Cov}(D, U)$. Derive the population regression slope of Y on D, and explain why it differs from the structural effect.

Solution 9: Group means and the omitted structural error

1. Take conditional expectations of the structural equation:

$$\mathbb{E}(Y \mid D = d) = 80 + 5d - 10\mathbb{E}(Z \mid D = d) + \mathbb{E}(V \mid D = d).$$

Hence

$$\mathbb{E}(Y \mid D = 0) = 80 - 10(1) = 70, \qquad \mathbb{E}(Y \mid D = 1) = 80 + 5 - 10(2) = 65.$$

The observed mean gap is $65 - 70 = \boxed{-5 \text{ points}}$.

2. With $U = -10Z + V$,

$$\mathbb{E}(U \mid D = d) = -10\mathbb{E}(Z \mid D = d) + \mathbb{E}(V \mid D = d),$$
$$\mathbb{E}(U \mid D = 0) = -10, \qquad \mathbb{E}(U \mid D = 1) = -20.$$

The structural error has nonzero conditional means that differ across tutoring groups. The zero-conditional-mean assumption fails.

Solution 9: The population slope and the structural effect

3. For binary D with $p = \Pr(D = 1)$, $\mathrm{Var}(D) = p(1-p) = .25$. Also,

$$\mathbb{E}(U) = \sum_{d=0}^{1} \mathbb{E}(U \mid D = d) \Pr(D = d) = .5(-10) + .5(-20) = -15,$$

$$\mathbb{E}(DU) = \Pr(D = 1)\mathbb{E}(U \mid D = 1) = .5(-20) = -10,$$

$$\mathrm{Cov}(D, U) = \mathbb{E}(DU) - \mathbb{E}(D)\mathbb{E}(U) = -10 - (.5)(-15) = -2.5.$$

From $Y = 80 + 5D + U$, the population projection slope is

$$\beta_1^* = \frac{\mathrm{Cov}(D, Y)}{\mathrm{Var}(D)} = 5 + \frac{\mathrm{Cov}(D, U)}{\mathrm{Var}(D)} = 5 + \frac{-2.5}{.25} = \boxed{-5}.$$

The structural effect is $+5$ points, but greater disadvantage among participants creates a negative observational gap.

The omitted structural error U differs from a projection error or a sample OLS residual. Orthogonality of OLS residuals does not establish structural exogeneity or causality.

Problem 10: One unusual score and repeated students

Return to Problem 2, with $x = (1, 2, 3, 4, 5)$ and $y = (52, 54, 53, 58, 58)$. Replace only the last score, $y_5 = 58$, by 90. Originally, $\bar{x} = 3$, $S_{XX} = 10$, and $\widehat{\beta}_1 = 1.6$.

1. Recalculate $\bar{y}$, S_{XY}, the OLS slope, and the intercept using the changed data.
2. Derive the change in the slope when only one y_j changes. Apply it here with $\Delta y_5 = 32$. What should we investigate before deciding how to handle this observation?
3. Compare two designs: 100 independent draws from the same student population, versus 20 weekly observations for each of five students. Is i.i.d. sampling justified equally in both? If hours and scores are bounded, what does that imply about fourth moments, independence, and influential observations?

Solution 10: Re-estimating after changing one score

1. The regressor values are unchanged, so $\bar{x} = 3$ and $S_{XX} = 10$. With the new score,

$$\bar{y}_{\text{new}} = \frac{1}{n}\sum_i y_{i,\text{new}} = \frac{52 + 54 + 53 + 58 + 90}{5} = \frac{307}{5} = 61.4.$$

$$\begin{aligned} S_{XY,\text{new}} &= \sum_i (x_i - \bar{x})(y_{i,\text{new}} - \bar{y}_{\text{new}}) \\ &= 18.8 + 7.4 + 0 - 3.4 + 57.2 = 80. \end{aligned}$$

The new coefficients are

$$\widehat{\beta}_{1,\text{new}} = \frac{S_{XY,\text{new}}}{S_{XX}} = \frac{80}{10} = \boxed{8},$$

$$\widehat{\beta}_{0,\text{new}} = \bar{y}_{\text{new}} - \widehat{\beta}_{1,\text{new}}\bar{x} = 61.4 - 8(3) = \boxed{37.4}.$$

Changing one score moves the fitted line from $50.2 + 1.6x$ to $37.4 + 8x$.

Solution 10: Influence and the unit of sampling (continued)

2. Since $\sum_i (x_i - \bar{x}) = 0$, write

$$\widehat{\beta}_1 = \frac{\sum_i (x_i - \bar{x}) y_i}{S_{xx}}.$$

With all x_i fixed and only y_j changed,

$$\Delta\widehat{\beta}_1 = \frac{(x_j - \bar{x})\Delta y_j}{S_{xx}} = \frac{(5-3)(90-58)}{10} = 6.4, \qquad 1.6 + 6.4 = 8.$$

Check recording, coding, units, and context. A valid extreme score should not be deleted automatically because it changes the line.

3. Independent draws from the same student population support the i.i.d. assumption at the student level. Repeated observations of five students may share individual factors and dependence over time; 100 records need not be 100 independent observations.

Bounded hours and scores have finite fourth moments. Boundedness does not guarantee independence or prevent a realized observation from strongly influencing a small-sample regression.

Problem 11: Are both OLS coefficients unbiased?

For the full regressor sample $\mathbf{X} = (1, 2, 3)$, outcomes satisfy

$$Y_i = 10 + 2X_i + U_i.$$

Conditional on the full $\mathbf{X}$, the errors are independent and each equals -1 or 1 with probability $1/2$.

1. Show that $\mathbb{E}[X_i U_i \mid \mathbf{X}] = X_i \mathbb{E}[U_i \mid \mathbf{X}] = 0$. Why can X_i be taken outside this expectation?
2. With $\overline{X} = 2$ and $S_{XX} = \sum_i (X_i - \overline{X})^2 = 2$, express each OLS coefficient as its true value plus a weighted sum of errors. Find their conditional means. How does iterated expectations extend the result to random regressors?
3. List the possible values of $\widehat{\beta}_1$, their probabilities, and their mean.

Solution 11: Fixed conditional weights

1. Given $\mathbf{X}$, each X_i is a known constant. Since

$$\mathbb{E}[U_i \mid \mathbf{X}] = (-1)\frac{1}{2} + (1)\frac{1}{2} = 0, \qquad \mathbb{E}[X_i U_i \mid \mathbf{X}] = X_i \mathbb{E}[U_i \mid \mathbf{X}] = \boxed{0}.$$

2. The OLS slope error is a weighted sum:

$$\widehat{\beta}_1 - \beta_1 = \frac{\sum_i (X_i - \overline{X}) U_i}{S_{XX}} = \frac{-U_1 + 0U_2 + U_3}{2},$$

$$\widehat{\beta}_1 = \boxed{2 + \frac{U_3 - U_1}{2}}.$$

The intercept uses $\overline{Y} = \beta_0 + \beta_1 \overline{X} + \overline{U}$:

$$\begin{aligned}\widehat{\beta}_0 - \beta_0 &= \overline{U} - \overline{X}(\widehat{\beta}_1 - \beta_1) \\ &= \frac{U_1 + U_2 + U_3}{3} - 2\frac{U_3 - U_1}{2} = \frac{4U_1 + U_2 - 2U_3}{3}.\end{aligned}$$

Thus $\boxed{\widehat{\beta}_0 = 10 + (4U_1 + U_2 - 2U_3)/3}$.

Solution 11: Unbiasedness and a slope distribution (continued)

2. Taking conditional expectations of the error expressions gives

$$\mathbb{E}[\widehat{\beta}_1 \mid \mathbf{X}] = 2 + \frac{\mathbb{E}[U_3 \mid \mathbf{X}] - \mathbb{E}[U_1 \mid \mathbf{X}]}{2} = \boxed{2},$$

$$\mathbb{E}[\widehat{\beta}_0 \mid \mathbf{X}] = 10 + \frac{4\mathbb{E}[U_1 \mid \mathbf{X}] + \mathbb{E}[U_2 \mid \mathbf{X}] - 2\mathbb{E}[U_3 \mid \mathbf{X}]}{3} = \boxed{10}.$$

If $\mathbf{X}$ is random, the same argument applies whenever $S_{XX} > 0$ and $\mathbb{E}[U_i \mid \mathbf{X}] = 0$. With finite coefficient expectations,

$$\mathbb{E}[\widehat{\beta}_j] = \mathbb{E}\Big[\mathbb{E}[\widehat{\beta}_j \mid \mathbf{X}]\Big] = \beta_j, \qquad j = 0, 1.$$

3. Independence makes the four (U_1, U_3) pairs equally likely:

$\widehat{\beta}_1$	1	2	3
Probability	1/4	1/2	1/4

$$\mathbb{E}[\widehat{\beta}_1 \mid \mathbf{X}] = \sum_b b \Pr(\widehat{\beta}_1 = b \mid \mathbf{X}) = 1\left(\frac{1}{4}\right) + 2\left(\frac{1}{2}\right) + 3\left(\frac{1}{4}\right) = 2.$$

Unbiasedness here is a finite-sample result; no large-n approximation is used.

Problem 12: How uncertain are the slope and intercept?

Use the design from Problem 2: $X = (1, 2, 3, 4, 5)$, $n = 5$, $\overline{X} = 3$, and $S_{XX} = 10$. In the population model $Y_i = \beta_0 + \beta_1 X_i + U_i$, assume, conditional on the full $\mathbf{X}$, mean-zero, uncorrelated errors with common variance $\sigma_U^2 = 4$.

1. Use the fixed slope weights to derive $\mathrm{Var}(\widehat{\beta}_1 \mid \mathbf{X})$, then calculate its theoretical standard deviation.
2. Write the intercept error using weights $a_i = 1/n - \overline{X}(X_i - \overline{X})/S_{XX}$. Derive its conditional variance and calculate its theoretical standard deviation.
3. Now treat σ_U^2 as unknown. Use Problem 7's $SSR = 6.4$ to estimate the standard errors of both coefficients. Distinguish these from the theoretical standard deviations above.

Solution 12: The slope's conditional variance

1. Let $d_i = X_i - \overline{X}$ and $w_i = d_i / S_{XX}$. Then

$$\widehat{\beta}_1 - \beta_1 = \sum_i w_i U_i, \qquad (w_1, w_2, w_3, w_4, w_5) = \frac{1}{10}(-2, -1, 0, 1, 2).$$

The weights are fixed given $\mathbf{X}$. Conditional cross covariances vanish:

$$\begin{aligned} \mathrm{Var}(\widehat{\beta}_1 \mid \mathbf{X}) &= \sum_i w_i^2 \, \mathrm{Var}(U_i \mid \mathbf{X}) = \sigma_U^2 \sum_i \frac{d_i^2}{S_{XX}^2} \\ &= \frac{\sigma_U^2 S_{XX}}{S_{XX}^2} = \frac{\sigma_U^2}{S_{XX}} = \frac{4}{10} = \boxed{.4}. \end{aligned}$$

Its theoretical standard deviation is the square root:

$$\mathrm{SD}(\widehat{\beta}_1 \mid \mathbf{X}) = \sqrt{\mathrm{Var}(\widehat{\beta}_1 \mid \mathbf{X})} = \sqrt{.4} \approx \boxed{.63246}.$$

These calculations use the population error variance $\sigma_U^2 = 4$.

Solution 12: The intercept's conditional variance (continued)

2. The intercept error can also be written as a fixed weighted sum:

$$\widehat{\beta}_0 - \beta_0 = \overline{U} - \overline{X}\frac{\sum_i d_i U_i}{S_{XX}} = \sum_i a_i U_i, \qquad a_i = \frac{1}{n} - \frac{\overline{X} d_i}{S_{XX}}.$$

Using $\sum_i d_i = 0$ and $\sum_i d_i^2 = S_{XX}$,

$$\sum_i a_i^2 = \frac{1}{n} - \frac{2\overline{X}}{nS_{XX}} \underbrace{\sum_i d_i}_{0} + \frac{\overline{X}^2}{S_{XX}^2} \underbrace{\sum_i d_i^2}_{S_{XX}}$$

$$= \frac{1}{n} + \frac{\overline{X}^2}{S_{XX}}.$$

Thus conditional uncorrelatedness gives

$$\mathrm{Var}(\widehat{\beta}_0 \mid \mathbf{X}) = \sigma_U^2 \sum_i a_i^2 = 4\left(\frac{1}{5} + \frac{9}{10}\right) = \boxed{4.4},$$

$$\mathrm{SD}(\widehat{\beta}_0 \mid \mathbf{X}) = \sqrt{4.4} \approx \boxed{2.09762}.$$

Solution 12: Estimating both standard errors (continued)

3. Replace the unknown population error variance by

$$\widehat{\sigma}_U^2 = SER^2 = \frac{SSR}{n-2} = \frac{6.4}{5-2} = \frac{32}{15}.$$

Take the square root of each estimated coefficient variance:

$$SE(\widehat{\beta}_1) = \sqrt{\frac{SER^2}{S_{XX}}} = \sqrt{\frac{32/15}{10}} \approx \boxed{.46188},$$

$$SE(\widehat{\beta}_0) = \sqrt{SER^2\left(\frac{1}{n} + \frac{\overline{X}^2}{S_{XX}}\right)}$$

$$= \sqrt{\frac{32}{15}\left(\frac{1}{5} + \frac{9}{10}\right)} \approx \boxed{1.53188}.$$

The theoretical SDs use the true $\sigma_U^2 = 4$; the estimated SEs use the sample's $SER^2 = 32/15$. The latter vary across samples and need not equal the theoretical SDs in a particular sample.

With only five observations, these variance calculations do not by themselves justify large-sample normal tests or intervals.

Problem 13: What makes a slope estimate precise?

Three fixed designs satisfy the mean-zero, homoskedastic, conditionally uncorrelated error assumptions, with $\sigma_U = 4$:

Design	n	$\overline{X}$	S_{XX}
A	100	0	100
B	100	0	400
C	400	0	400

1. Calculate the theoretical conditional SDs of both OLS coefficients in each design. Explain the roles of regressor variation and sample size.
2. How do all these SDs change if σ_U doubles to 8?
3. A fourth design has $n = 100$, $\overline{X} = 10$, and $S_{XX} = 100$. Calculate both SDs. Then recenter with $Z = X - 10$ and calculate the new intercept's SD. What changes in the meaning of the intercept?

Solution 13: Regressor spread, sample size, and noise

1. Take square roots of the conditional coefficient variances:

$$\mathrm{SD}(\widehat{\beta}_1 \mid \mathbf{X}) = \sqrt{\frac{\sigma_U^2}{S_{XX}}} = \frac{\sigma_U}{\sqrt{S_{XX}}}, \qquad \mathrm{SD}(\widehat{\beta}_0 \mid \mathbf{X}) = \sigma_U \sqrt{\frac{1}{n} + \frac{\overline{X}^2}{S_{XX}}}.$$

Design	Slope SD	Intercept SD
A	$4/\sqrt{100} = \boxed{.4}$	$4\sqrt{1/100} = \boxed{.4}$
B	$4/\sqrt{400} = \boxed{.2}$	$4\sqrt{1/100} = \boxed{.4}$
C	$4/\sqrt{400} = \boxed{.2}$	$4\sqrt{1/400} = \boxed{.2}$

From A to B, more variation in X halves the slope SD. From B to C, S_{XX} stays fixed, so the slope SD stays fixed despite the larger n. With $\overline{X} = 0$, the intercept SD falls as $1/\sqrt{n}$.

2. Both SD formulas are proportional to σ_U. Doubling σ_U to 8 doubles every SD; it multiplies every variance by four.

Solution 13: Where is the intercept measured? (continued)

3. For $n = 100$, $\overline{X} = 10$, and $S_{XX} = 100$,

$$\mathrm{SD}(\widehat{\beta}_1 \mid \mathbf{X}) = \sqrt{\frac{16}{100}} = \boxed{.4},$$

$$\mathrm{SD}(\widehat{\beta}_0 \mid \mathbf{X}) = \sqrt{16\left(\frac{1}{100} + \frac{10^2}{100}\right)} = \sqrt{16.16} \approx \boxed{4.020}.$$

If $Z = X - 10$, then $\overline{Z} = 0$ and $S_{ZZ} = S_{XX} = 100$. Write

$$Y = \gamma_0 + \beta_1 Z + U, \qquad \gamma_0 = \beta_0 + 10\beta_1.$$

The new intercept estimates the line at old $X = 10$, rather than old $X = 0$:

$$\mathrm{SD}(\widehat{\gamma}_0 \mid \mathbf{X}) = \sqrt{16\left(\frac{1}{100} + \frac{0^2}{100}\right)} = \boxed{.4}.$$

The slope and fitted line are unchanged. Recentring changes the intercept's target; it does not improve the precision of the fitted line at any given X.

Problem 14: Is the intercept or slope different from zero?

Regress test-score gains Y (points) on weekly tutoring hours X. A large i.i.d. sample gives

$$n = 102, \quad \overline{X} = 10, \quad S_{XX} = 400, \quad SSR = 400, \qquad \widehat{\beta}_0 = 1.5, \quad \widehat{\beta}_1 = .3.$$

Assume zero conditional-mean errors, homoskedasticity, conditionally uncorrelated errors, finite fourth moments, and a valid large-sample normal approximation.

1. Calculate SER and the estimated SEs of both coefficients.
2. State a two-sided zero-null test for each coefficient and compute each observed test statistic.
3. Find both approximate p-values and give the 5% decisions. Use $\Phi(3) = .99865$ and $\Phi(1.4714) \approx .9294$.
4. Construct a 95% confidence interval for each coefficient and check whether it contains zero. Are these two separate tests a joint test of both coefficients?

Solution 14: Estimate the error variance and both SEs

1. With an intercept and one slope, the error variance estimate uses $n - 2$:

$$SER^2 = \frac{SSR}{n-2} = \frac{400}{102-2} = 4, \qquad SER = \sqrt{\frac{SSR}{n-2}} = \sqrt{4} = \boxed{2} \text{ points.}$$

Each coefficient SE is the square root of its estimated variance:

$$SE(\widehat{\beta}_1) = \sqrt{\frac{SER^2}{S_{XX}}} = \sqrt{\frac{4}{400}} = \sqrt{.01} = \boxed{.1},$$

$$SE(\widehat{\beta}_0) = \sqrt{SER^2\left(\frac{1}{n} + \frac{\overline{X}^2}{S_{XX}}\right)}$$

$$= \sqrt{4\left(\frac{1}{102} + \frac{100}{400}\right)} \approx \boxed{1.01942}.$$

The slope SE is measured in points per weekly tutoring hour; the intercept SE is measured in points.

Solution 14: Two separate zero-null tests (continued)

2. For each coefficient j, the zero-null statistic is

$$t_j^{\text{obs}} = \frac{\widehat{\beta}_j - 0}{SE(\widehat{\beta}_j)}.$$

Coefficient	Hypotheses	Observed statistic
Slope	$H_0 : \beta_1 = 0$ vs. $H_1 : \beta_1 \neq 0$	$.3/.1 = \boxed{3}$
Intercept	$H_0 : \beta_0 = 0$ vs. $H_1 : \beta_0 \neq 0$	$1.5/1.01942 \approx \boxed{1.4714}$

3. Under its null, use the large-sample normal approximation:

$$p_j \approx 2[1 - \Phi(|t_j^{\text{obs}}|)].$$

$$p_1 \approx 2(1 - .99865) = \boxed{.0027} < .05 \quad \Longrightarrow \text{ reject } \beta_1 = 0,$$

$$p_0 \approx 2(1 - .9294) = \boxed{.1412} > .05 \quad \Longrightarrow \text{ do not reject } \beta_0 = 0.$$

The slope is statistically different from zero at 5%; the intercept is not.

Solution 14: Confidence intervals (continued)

4. For each coefficient separately, the 95% normal approximate interval is

$$CI_{.95}(\beta_j) = [\widehat{\beta}_j - 1.96SE(\widehat{\beta}_j),\ \widehat{\beta}_j + 1.96SE(\widehat{\beta}_j)].$$

For the slope,

$$.3 \pm 1.96(.1) = \boxed{[.104,\ .496]}.$$

For the intercept, using its unrounded SE,

$$1.5 \pm 1.96\sqrt{4\left(\frac{1}{102} + \frac{100}{400}\right)} \approx \boxed{[-.498,\ 3.498]}.$$

Zero lies outside the slope interval and inside the intercept interval, matching the separate 5% tests.

Separate coefficient inference

These are two separate tests and two marginal 95% intervals. They do not provide a joint test that both coefficients are zero or a simultaneous 95% confidence region.